

Shehab Ahmed Taia

Mechatronics Engineering MEng Student

University of Birmingham

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Education

Year 2 – MEng Mechatronics Engineering

University of Birmingham

Expected Graduation: 2028

Relevant Modules: Digital Electronics, Engineering Mathematics, Mechanics, Electrical Machines, Electromagnetism, Control, Microprocessors, Analogue Devices

Year 1 (Pathway) – Mechatronics Engineering

Kaplan International College London

Academic Year: 2024-2025

Relevant Modules: Engineering Mathematics, Mechanics, Thermodynamics, Thermofluids, Research Design Development, Programming

Skills

Languages: C/C++ (Arduino), Python, MATLAB

Software & Tools: Simulink, Simscape Electrical, Multisim Live, TinkerCAD, AutoCAD, LaTeX, Microsoft Excel

Hardware: Circuit prototyping, breadboard assembly, logic gate design, embedded systems

Engineering: Power electronics simulation, control systems modelling, finite state machine design, technical documentation

Projects

Underwater Habitat Electrical Control System – IDP2

Jan 2026 – May 2026

Individual design project (Systems & Control work package)

- Designed a **Simscape Electrical** model in MATLAB/Simulink simulating a **48V DC** power distribution system with a 2:1 step-down converter supplying 6+ subsystem loads across 3 operating modes (Normal, Docking, Emergency)
- Implemented closed-loop fault detection using **MATLAB Function blocks**, achieving automatic branch isolation and load shedding that reduced Emergency mode power draw from 125W to 74W, protecting critical systems
- Modelled all loads (SCADA, communications, sensors, data logger, docking charger, electromagnetic clamps) as **variable resistors with thermal mass**, simulating real-world power fluctuations and temperature effects
- Maintained system efficiency **above 92.8% across all operating modes**, exceeding 97% in Normal and Emergency modes by minimising resistive losses through optimised load resistance values
- Designed SCADA-based control architecture and a hybrid optical/acoustic underwater communications network in the concept design phase, covering 8 subsystems across the full habitat

Autonomous Fire-Fighting Vehicle

Jan 2025 – May 2025

Six-member team project

- Played a leading role in the technical development of an autonomous fire-fighting vehicle using an **Arduino Uno R3**
- The system was built and tested in **TinkerCAD**, simulating 13+ components
- Wrote Arduino code from scratch integrating 4 independent subsystems: wheel motor control, ultrasonic distance sensing, infrared fire detection, and water pump activation
- Debugged power supply and motor control issues to ensure reliable autonomous operation

- Delivered an operating system capable of avoiding obstacles, locating a fire source, and extinguishing fires, validated through repeated test scenarios

Digital Electronics – Hardware Finite State Machine

Oct 2025 – Dec 2025

Two-member team project

- Designed a synchronous finite state machine with **13 specified states**, including handling unallocated states
- Collaboratively developed state diagrams, transition tables, and Karnaugh maps to ensure logical correctness
- Implemented and validated the logic in **Multisim Live**, followed by breadboard layout and wiring planning in **TinkerCAD**
- Built and tested the physical circuit using **7474 D flip-flops**, clock logic, and **7400/7410/7420 NAND gates**
- Documented the full design and testing process in a structured engineering logbook with 30+ pages of structured documents

Work Experience

IT Department Internship: Talaat Mostafa Group

Dec 2023

- Completed an enterprise IT placement at one of Egypt's largest conglomerates, gaining hands-on exposure to server environments, networking infrastructure, and disk management operations